
1. INTRODUCTION: GETTING STARTED

1.1 Setup and Overview

Welcome to the second half of this course: the world of quantum computing! This worksheet will guide us through fundamental quantum gates using the Quantum Computer Gate Playground.

What You'll Learn: By the end of this exercise, you'll be able to describe the effects of different quantum gates on a single qubit and visualize quantum superposition and interference.

Required Links:

- Background Reading: <https://davidbkemp.github.io/QuantumComputingArticle/>
- Interactive Playground: <https://davidbkemp.github.io/quantum-gate-playground/>

Before You Begin: Open the Quantum Computer Gate Playground link above. You should see a visualization with a blue disc representing the quantum state of a qubit. This disc can move around as we apply different gates!

A Quick Reminder About Qubits: Unlike classical bits (which are either 0 or 1), qubits can exist in a **superposition** of both states simultaneously:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here, α and β are complex numbers called probability amplitudes. The probabilities are given by $|\alpha|^2$ for state $|0\rangle$ and $|\beta|^2$ for state $|1\rangle$.

2. SINGLE QUBIT GATES

2.1 The Hadamard Gate (H)

The Hadamard gate is one of the most important gates in quantum computing. It creates superposition from definite states!

Activity:

1. In the playground, select the Hadamard gate from the dropdown menu
2. Apply the Hadamard gate to the qubit (which starts in state $|0\rangle$)
3. Carefully observe what happens to the blue disc

Questions to Answer:

- 1a.** What is the qubit's new state after applying the Hadamard gate? (Hint: Look at the coefficients displayed for $|0\rangle$ and $|1\rangle$)
- 1b.** What do you notice about the magnitude and phase of the state? Are the probabilities equal?
- 1c.** In your own words, explain what the Hadamard gate does. Why is this gate so important for quantum computing?

Mathematical Insight: The Hadamard gate transforms states as follows:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Notice how it creates an equal superposition, but with different relative phases!

2.2 The X Gate (NOT Gate)

The X gate is the quantum version of the classical NOT gate. Let's explore its behavior.

Activity:

- 1.** Reset the qubit to $|0\rangle$ (refresh the page if needed)
- 2.** Select and apply the X (NOT) gate
- 3.** Observe the change in state

Questions to Answer:

- 2a.** What state is the qubit in after applying the X gate once? Apply the X gate again. What happens?
- 2b.** How does this gate compare to the classical NOT operation? Are they the same or different?

Mathematical Insight: The X gate acts like a bit flip:

$$X|0\rangle = |1\rangle \quad \text{and} \quad X|1\rangle = |0\rangle$$

This is exactly like flipping a classical bit! However, unlike classical gates, quantum gates must be **reversible**. Notice that applying X twice returns you to the original state.

2.3 The Z Gate (Phase Flip)

The Z gate doesn't flip between $|0\rangle$ and $|1\rangle$, but instead changes the **relative phase** between them. This is a purely quantum effect!

Activity:

- 1.** Start fresh with $|0\rangle$

2. Apply a Hadamard gate first (to create superposition)
3. Then apply a Z gate
4. Carefully observe how the phase changes

Questions to Answer:

3a. What does the Z gate do to the quantum state? How does the visualization change?

3b. How does the effect of the Z gate differ from the X gate? Why might we need both?

Important Concept: The Z gate leaves $|0\rangle$ unchanged but adds a phase of -1 to $|1\rangle$:

$$Z|0\rangle = |0\rangle \quad \text{and} \quad Z|1\rangle = -|1\rangle$$

On a state like $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, it produces $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. The **global** probabilities remain the same, but the **relative** phase changes. This phase difference is crucial for quantum interference!

2.4 The Y Gate

The Y gate combines aspects of both X and Z gates. It performs a bit flip *and* a phase flip!

Activity:

1. Reset to $|0\rangle$
2. Apply the Y gate
3. Observe and record the effect on the state vector

Questions to Answer:

4a. Describe the effect of the Y gate on the state vector. What changes do you see?

4b. How does this compare to the X and Z gates? Can you think of the Y gate as a combination of the other two?

Mathematical Note: The Y gate performs:

$$Y|0\rangle = i|1\rangle \quad \text{and} \quad Y|1\rangle = -i|0\rangle$$

Notice the imaginary unit i appearing! This is another purely quantum feature. The Y gate is actually equivalent to applying X, then Z, then multiplying by i .

3. ROTATION GATES

3.1 Continuous Rotations

So far, we've seen gates that create specific fixed transformations. But quantum gates can also perform **continuous rotations** around different axes of the Bloch sphere!

Background: The X, Y, and Z gates we've seen are actually special cases of rotation gates rotated by π radians (180°). General rotation gates let us rotate by any angle!

Activity: X, Y, and Z Rotations

1. Start with $|0\rangle$
2. Apply a $\pi/4$ (45-degree) rotation around the X-axis
3. Observe how the amplitude and phase change
4. Reset and repeat for Y rotation and Z rotation at the same angle

Questions to Answer:

- 5a.** Compare the effects of $\pi/4$ rotations around each axis (X, Y, and Z). How are they similar? How are they different?
- 5b.** How do these rotation gates differ from the standard X, Y, and Z gates you explored earlier? (Hint: Think about the angle of rotation)
- 5c.** What do you think would happen if you applied a $\pi/8$ rotation around an axis, and then applied it again? (Try it!)

Advanced Insight: The rotation gates are written as:

$$R_x(\theta) = \begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

And similarly for $R_y(\theta)$ and $R_z(\theta)$. When $\theta = \pi$, these reduce to the X, Y, and Z gates! This means any quantum gate on a single qubit can be built from rotations.

4. ADVANCED OPERATIONS

4.1 Quantum Fourier Transform (QFT)

The Quantum Fourier Transform is a powerful operation used in many quantum algorithms, including Shor's algorithm for factoring numbers!

Activity:

1. Reset to $|0\rangle$
2. Apply the QFT gate
3. Observe and describe the effect on the state

Questions to Answer:

6a. Describe what happens to the quantum state when you apply the QFT. How does it differ from the Hadamard gate?

6b. Based on what you've observed, how do you think the QFT might be useful in quantum algorithms? What kind of problems might benefit from this transformation?

Note: For a single qubit, the QFT is actually identical to the Hadamard gate! But for multiple qubits, the QFT becomes much more complex and powerful. It can find periodicities in data, which is crucial for algorithms like Shor's factoring algorithm and quantum phase estimation.

5. MEASUREMENT AND PROBABILITY

5.1 The Role of Measurement

One of the most surprising aspects of quantum mechanics is that **measurement changes the quantum state!** This is called **wavefunction collapse**.

Activity: Multiple Measurements

1. Apply a Hadamard gate to create a superposition
2. Measure the qubit and record the outcome (0 or 1)
3. Reset and repeat this process at least 10 times
4. Keep track of how many times you get 0 versus 1

Questions to Answer:

7a. After applying the Hadamard gate and measuring multiple times, what do you notice about the probabilities of getting 0 versus 1? Are they equal?

7b. Before you measure, the qubit is in a superposition. What happens to this superposition when you measure? Can you measure the same qubit twice and get different results?

7c. How does measurement affect the quantum state? Is this effect reversible?

Key Concept: When you measure a qubit in the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

- You get outcome 0 with probability $|\alpha|^2$
- You get outcome 1 with probability $|\beta|^2$
- After measurement, the superposition **collapses** to either $|0\rangle$ or $|1\rangle$
- If you measure again immediately, you'll get the same result!

This irreversibility of measurement is one of the deepest mysteries of quantum mechanics.

6. REFLECTION AND SYNTHESIS

6.1 Putting It All Together

Now that you've explored various quantum gates, let's reflect on what makes quantum computing unique and powerful.

8. Which gates create superposition? List all the gates you've encountered that can take a definite state like $|0\rangle$ and create a superposition.

9. Which gates change only the relative phase of a qubit without affecting the probability of measuring 0 or 1? Why might these "phase gates" be important even though they don't change measurement probabilities?

10. Does any of this surprise you? Is this how you thought a quantum computer would work? What's the most counterintuitive thing you've learned?

11. Challenge Question: Suppose you want to create the state $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ starting from $|0\rangle$. What sequence of gates would you apply? (Hint: You'll need at least two gates.)

12. Real-World Connection: Quantum computers can solve certain problems exponentially faster than classical computers. Based on what you've learned about superposition and gates, why do you think quantum computers might be more powerful for some tasks? What kinds of problems do you think would benefit most from quantum computing?

Final Thought: The gates you've explored today are the building blocks of all quantum algorithms. Companies like IBM, Google, and others are building quantum computers with tens to thousands of qubits. While we're still in the early stages, quantum computers may one day revolutionize fields like cryptography, drug discovery, optimization, and artificial intelligence!